CREDIT CARD FRAUD DETECTION VIA RANDOM UNDERSAMPLING, SMOTE OVERSAMPLING, AND PENALIZED LOGISTIC REGRESSION

Aaheri Gangopadhyay

Repository: github.com/aaherigangopadhyaystats-code

ABSTRACT

Credit card fraud detection is challenging because legitimate transactions greatly outnumber fraudulent ones. This project analyzes a reported dataset of 23,323 transaction records and compares two approaches to class imbalance: random undersampling and the Synthetic Minority Over-sampling Technique (SMOTE). Logistic Regression trained on an undersampled subset achieved 98.27% training accuracy and 96.55% test accuracy. An L2-penalized Logistic Regression model trained on SMOTE-balanced training data was evaluated on a held-out test set of 4,665 transactions, retaining the natural class distribution for testing. The model achieved 99.00% overall test accuracy and 100.00% recall for the 17 fraud cases in the test set, with 25.00% precision for the fraud class. The results illustrate why recall and precision should be considered alongside accuracy when evaluating classifiers under severe class imbalance.

1. PROJECT OVERVIEW AND RATIONALE

Fraud-detection systems must balance missed fraudulent transactions against false alarms on legitimate purchases. Severe class imbalance can make overall accuracy misleading, particularly when fraudulent transactions represent only a small fraction of all records. The analysis compares random undersampling with SMOTE-based oversampling and emphasizes recall and precision-recall performance as practical measures for fraud classification.

2. DATASET AND EXPLORATORY ANALYSIS

The report describes a financial transaction dataset with numerical predictors and a binary target label (Class: 0 = legitimate, 1 = fraudulent). The predictors include elapsed time, 28 anonymized PCA-transformed features (V1–V28), and transaction amount.

- **Reported total:** 23,323 transaction records.
- **Reported class counts:** 23,235 legitimate transactions (99.62%) and 87 fraudulent transactions (0.38%).
- **Data cleaning:** One incomplete row in V23–V28 and Amount was reported as dropped before modeling.

Exploratory comparisons reported a mean transaction amount of $73.34 (median $17.99) for legitimate transactions and $92.92 (median $1.00) for fraudulent transactions. Fraudulent transaction amounts ranged from approximately $0–$1 test authorizations to a reported maximum of $1,809.68. The mean of feature V3 was reported as +0.78 for legitimate transactions and –12.34 for fraud; feature V4 averaged +0.23 for legitimate transactions and +6.27 for fraud.

3. PREPROCESSING AND RESAMPLING

3.1 Random undersampling

- The report states that all 87 fraud records were isolated.
- It reports selecting 492 legitimate records to form a balanced subset of 579 total rows.
- The subset was split into an 80% training set (n = 463) and a 20% test set (n = 116).

3.2 SMOTE on the training partition

- The cleaned data were split into 80% training (n = 18,657) and 20% testing (n = 4,665).
- SMOTE was applied only to the training split, with the training classes resampled to a 1:1 balance.
- The test partition retained the reported natural class distribution to represent evaluation on imbalanced transactions.

The preprocessing description emphasizes separating the held-out test data from the resampling process to reduce the risk of data leakage.

4. MODELING EXPERIMENTS AND FINDINGS

4.1 Logistic Regression on undersampled data

A standard Logistic Regression model was fitted to the reported balanced undersampled subset.

- Training accuracy: 98.27% (455/463 reported correct predictions).
- Test accuracy: 96.55% (112/116 reported correct predictions).

The report notes that discarding a large share of legitimate transactions can reduce the information available for training and may increase false alarms on unseen legitimate transactions.

4.2 SMOTE with L2-penalized Logistic Regression

An L2-penalized Logistic Regression model was trained on the SMOTE-balanced training partition with penalty = 'l2' and max_iter = 1000. Evaluation used a held-out test set of 4,665 transactions, reported as 4,648 legitimate and 17 fraudulent.

5. HELD-OUT TEST PERFORMANCE

Metric	Legitimate class (0)	Fraud class (1)
Precision	100.00%	25.00%
Recall	99.00%	100.00%
F1-score	0.99	0.40
Support	4,648	17

- **Overall test accuracy:** 99.00%.
- **Fraud recall:** 100.00%; all 17 fraud cases in the test set were reported as detected.
- **Fraud precision:** 25.00%; at this operating point, a substantial proportion of transactions flagged as fraud were legitimate and would require review.

The reported 100.00% fraud recall is based on 17 fraud cases in the test set and should be interpreted with that limited support in mind. Precision and the costs of false positives are important when choosing an operational threshold.

6. CONCLUSIONS AND FUTURE WORK

- **Accuracy can be deceptive:** A classifier that labels every transaction as legitimate could attain high accuracy under the reported class imbalance while failing to identify fraud.
- **Resampling strategy matters:** SMOTE was applied to the training partition while the test partition retained the natural class distribution.
- **Future work:** Evaluate non-linear ensemble methods such as XGBoost and Isolation Forest, and tune probability thresholds to balance precision and recall for different business costs.

The results provide an exploratory comparison of resampling strategies for fraud classification. Further validation, including repeated or time-aware evaluation and explicit review of false-positive costs, would be needed before drawing conclusions about performance in a live transaction system.